

Robot-Assisted Language Learning: A Meta-Analysis

Ali Derakhshan 
Golestan University

Timothy Teo
The Chinese University of Hong Kong

Esmael Saeedy Robat 
Islamic Azad University, Taybad Branch, Iran

Mostafa Janebi Enayati 
University of Maragheh, Iran

Akbar A. Jahanbakhsh
Ministry of Education, Iran

This meta-analysis included 27 empirical studies to explore the effectiveness of robot-assisted language learning (RALL) as well as the moderating effects of a number of variables. The comparison of 64 effect sizes from 2,637 participants indicated significant variability between the independent samples. The overall results showed that the combined linguistic and affective gain of RALL was significant. The results of moderator analysis showed that most moderators, including place of the study, type of robot, publication type, instructional level of the learners, design of the study, time points of assessment, learning setting, second language (L2) proficiency level, type of assessment, language domain, and the non-RALL condition had no significant effects on learning outcomes. However, the moderating effect of the participants' age was significant as RALL was found to be more effective for elementary school students (7–12 years old). The findings provide insights into the design of effective robots for second language acquisition (SLA) that support learners with certain characteristics.

KEYWORDS: robot-assisted language learning, second language acquisition, design-related factors, publication-related factors, participant-related factors, meta-analysis

Technologies such as interactive whiteboards (e.g., Teng, 2021), mobile applications (e.g., Lin & Lin, 2019), adventure video games (e.g., H. J. H. Chen & Yang, 2013; Janebi Enayat & Haghighatpasand, 2019), virtual reality (VR) headsets (e.g., Tai et al., 2020; Xie et al., 2021), and augmented reality (AR) displays (e.g., Teo et al., 2022; M.-H. Wu, 2021) are increasingly being incorporated into second language acquisition (SLA). The recent shift from conventional methods of language teaching and learning to the application of such newer forms of instruction has allowed a one-on-one learning environment (H. L. Chen et al., 2020) and ubiquitous learning (Dashtestani, 2016). The adoption of these technologies has also made it possible to provide tailored feedback to students, enabling educators to address their specific learning needs and promote their overall academic growth (Golonka et al., 2014). Moreover, these tools have the advantage of real-time interactivity, which fosters active participation among students, leading to positive attitudes and increased engagement (Y. Chen et al., 2020).

The significance of interaction has been recently acknowledged in computer-assisted language learning (CALL) research, which has led to the development of new means of instruction that enable language learners to communicate with technologies powered by language processing systems, deep learning techniques, and large-scale datasets (Fryer et al., 2019; Zeng et al., 2020).

One of the newest forms of technology following the crucial element of interactivity for learning a foreign language is robots, including software applications like chatbots that imitate human conversations (Huang et al., 2017; Yin & Satar, 2020), humanoid robots (Alemi & Bahramipour, 2019), social robots designed to interact with people (Alemi et al., 2015; van den Berghe et al., 2019), and intelligent personal assistants—smart speakers like Amazon Echo and Apple's HomePod that respond to voice commands (Moussalli & Cardoso, 2020). Such robots are thus different from robotic arms used for industrial purposes, often used to perform specific factory tasks. Robots have recently gained momentum in SLA research as they can potentially solve problems students often encounter, such as the cost of having tutors for conversation practice (Iio et al., 2018), anxiety about talking about unfamiliar topics (MacWhinnie & Mitchell, 2017), and lack of individualized feedback in traditional classes (Jiang & Yu, 2020).

Robots have certain advantages for language learning that are often missing in the other forms of technology. Social robots, for instance, “allow the learner to interact with the real-life physical environment” (van den Berghe et al., 2019, p. 260), which is essential for second language development (Alemi et al., 2014a; S. Lee et al., 2011). Second, they are adaptable and can use their sensors to detect the educational needs of language learners (Kanero et al., 2018). Additionally, physical robots have been reported to enhance the language skills and subskills of vocabulary (Ahmad et al., 2019; Alemi & Bahramipour, 2019; Alemi et al., 2015; de Wit et al., 2018; Demir-Lira et al., 2020; van den Berghe et al., 2021; Wedenborn et al., 2019), listening and reading (Hong et al., 2016), and speaking (Iio et al., 2018; Moussalli & Cardoso, 2017; Shin & Shin, 2015). Finally, social robots can not only reduce the anxiety of second language (L2) learners but also boost their motivation and engagement in the long run (Ahmad et al., 2019; Alemi et al., 2014b; de Haas et al., 2020; Hong et al., 2016; Hung et al., 2013; S. Lee et al., 2011).

Nonphysical robots, known as virtual software agents or bots, are likewise beneficial for SLA as they can be programmed electronically, process data, operate autonomously or semiautonomously, and exhibit intelligent behavior (H. L. Chen et al., 2020). Chatbots, for example, are now utilized for language learning purposes as they provide virtual communicative settings for the learners (H. L. Chen et al., 2020; Huang et al., 2017) and stimulate the learners' interest (Fryer et al., 2019). However, in a survey comparing physical robots and virtual agents, Li (2015) found that people interacted differently when the robots were physically present. The presence of robots resulted in persuasion, positive perceptions, and better user performance. More specifically, after reviewing previous studies on the application of social robots for education, Belpaeme et al. (2018) concluded that the physical presence of social robots has the advantage of enhancing the cognitive and affective dimensions of educational outcomes, whereas traditional technologies like on-screen displays do not yield the same results.

Despite the above-mentioned merits of robots for education, some studies have reported that robot-assisted language learning (RALL) has a number of challenges. Randall (2019), for instance, conducted a survey and mentioned the shortcomings of automatic speech recognition, such as problems in identifying word boundaries in continuous speech, speaker variations, and multimodal understanding of human speech as potential problems for RALL. In a systematic review, van den Berghe et al. (2019) also found conflicting results with respect to the social behavior of robots; some studies reported positive effects of social behavior on learning gains, and others found that this feature could be distracting and intimidating for young learners.

The contribution of robots to language learning, however, varied across studies, with the effect sizes ranging from small to large. These large variations may be attributed to a number of factors, such as design-related factors (e.g., using true or quasi-experimental research design, robot type, learning and teaching setting, dependent variables investigated, target language, and type of assessment), and participant-related factors (e.g., age, language proficiency, educational level, and country). This study aimed to investigate both by undertaking a meta-analysis. No previous meta-analysis has been conducted to address the possible justifications for variations seen across RALL studies; however, in a systematic review of 33 studies, van den Berghe et al. (2019) investigated the potential of robots as language tutors and found that RALL could yield different results depending on the age groups of the students and the target aspects of language. A meta-analysis has the advantages of extracting one effect size from the combined effect sizes of various quantitative studies with different sample sizes and exploring the more detailed relationships that might exist in the data (Bernard et al., 2004).

The extent to which variables related to design, publication, and participants could moderate the effect sizes of RALL studies has not yet been investigated in a meta-analytic study. The current investigation, therefore, set out to scrutinize these factors by undertaking a meta-analysis, which can provide viable answers to research questions that cannot be answered by conducting one single empirical study. A quantitative synthesis or meta-analysis is based on combining studies with different sample sizes to extract an effect size from all other studies (Pigott & Polanin, 2020). Therefore, it represents an alternative to systematic reviews that report results from studies with different sample sizes. It is "an approach to

estimating how much one treatment differs from another, over a large set of similar studies, along with the associated variability” (Bernard et al., 2004, p. 384). Another advantage of meta-analysis is the investigation of moderator variables to probe more detailed associations that might exist in the data. The following questions were thus addressed to accomplish the objectives of this study:

Research Question 1: What are the overall effects of RALL on linguistic and affective gains?

Research Question 2: Which moderator variables affect the overall effectiveness of RALL?

Method

Literature Search

As seen in Figure 1, PRISMA (preferred reporting items for systematic reviews and meta-analyses) provides pooled data and a comprehensive guideline for different sections of the systematic review report (Page et al., 2021). The data collection section, accordingly, comprises three main steps: identification (and collection) of the sources, screening of the sources to exclude duplicate/unrelated sources to obtain an eligible pool, and inclusion of the final sample. Following these steps, we thoroughly searched the electronic databases with the following search terms and phrases both separately and in combinations:

“Robot-assisted language teaching and learning,” “robot-aided language learning and teaching,” “robot(s) and language,” “RALL,” “artificial intelligence and language learning and/or teaching and learning,” “robots and English learning/teaching,” and “learning vocabulary with robots.” We conducted a comprehensive search of the research literature published before June 2021 using Scopus, Web of Science, SpringerLink, Wiley Online Library, ResearchGate, Oxford Journals, ERIC - Education Resources Information Center, ScienceDirect, Sage Journals Online, ProQuest, Cambridge Core, Google Scholar, JSTOR, ProQuest Dissertations & Theses Global, PsycINFO, Microsoft Academic Search and Linguistics, and Language Behavior Abstracts (LLBA). In addition, the reference sections of publications were also checked for more relevant studies. In the next phase of the study, we checked the ProQuest database to find unpublished doctoral dissertations.

Each study had to meet the following criteria to be included in this meta-analysis:

1. The studies investigated the effects of physical robots’ intervention on language learning outcomes and/or affective (motivation and/or confidence) gains.
2. The studies were reported in journals, conference proceedings, theses, or dissertations in English from 2010 to 2021.
3. The study design was either experimental or quasi-experimental.
4. Studies reported sufficient statistical information to calculate an effect size.

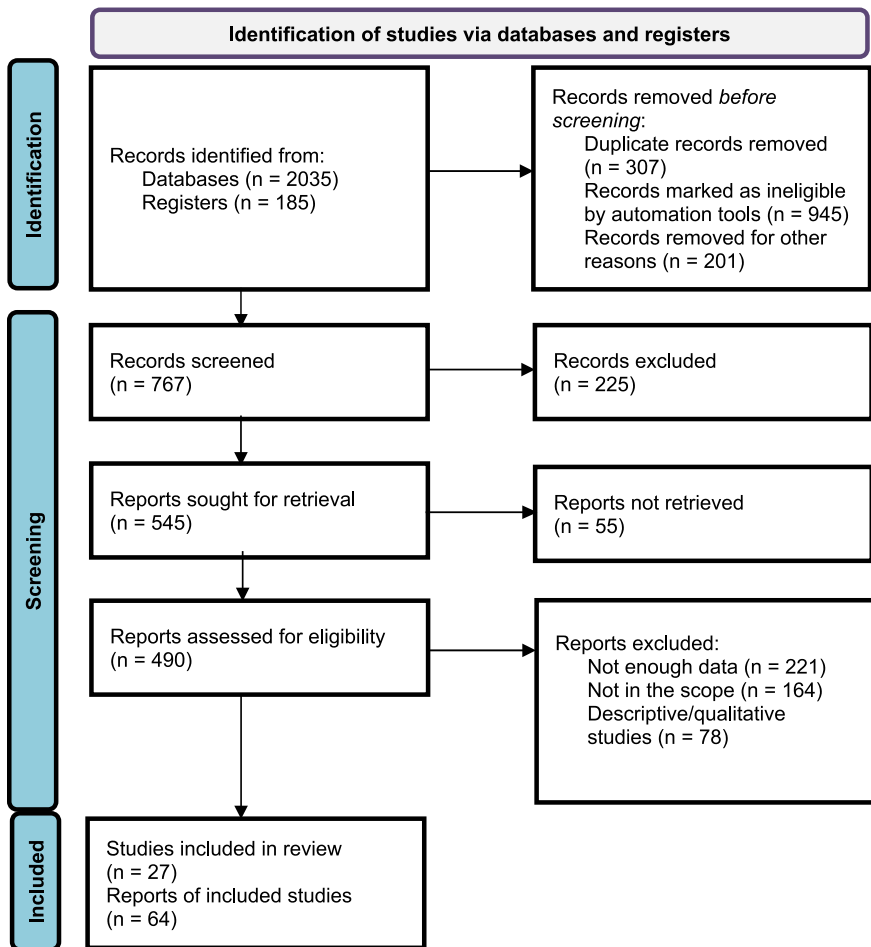


FIGURE 1. Study retrieval process using PRISMA (Page et al., 2021).

Studies were excluded if:

1. The robots were simply AR and VR agents.
2. They did not report enough quantitative data.
3. Their full texts were not available for consultation.
4. They did not include the impacts of robots on linguistic or affective gains of language learners.

Coding of Study Characteristics and Moderator Variables

For study codification, rating, and data management, the researchers employed Rayyan (www.rayyan.ai), a web and mobile application for systematic reviews

powered by artificial intelligence. Following data search and collection, all papers, along with bibliographic information, were exported to Rayyan. The platform provides the researchers with collaboration tools for blind rating and smart screening of papers. After multiple phases of title, abstract, and full-text screening, 255 papers were blindly rated by a team of four researchers. The initial screening of the abstracts reduced the number of articles to 65 based on the inclusion and exclusion criteria. Then, the remaining studies were more deeply screened by the third and fifth authors to meet the requirements of secondary analysis, based on which the authors reached an agreement for the inclusion of 39 papers and a conflict for 3 (Cohen's Kappa = .91). In a subsequent meeting, the authors discussed the three sources of conflict and reached a unanimous decision to exclude all three. Going through the reports in these papers, it was realized that 12 papers did not provide thorough statistics for the secondary analysis. We contacted the 12 authors and received relevant data for one more study. The team met again in cases of discrepancies and tried to make the inclusion criteria as transparent and explicit as possible. The consensus was established unanimously on the final inclusion of 27 studies for the analysis (see Table 1 for a complete list of included studies and their moderator variables). The final selected papers were coded by two of the four authors, and 64 effect sizes were drawn. The initial intercoder reliability was $\kappa = .884$, which reached a perfect agreement after double-checking the codes in a subsequent meeting.

A coding scheme was implemented using the previous review studies to search for and identify the RALL features from the 27 included samples that can be compared in this meta-analysis. For example, Randall (2019) analyzed the experimental studies based on features like the robots' form, function, and voice; the language skills taught by the robots; the affective results of using RALL; the age and language proficiency of the participants; and location. Additionally, van den Bergh et al. (2019) reviewed the previous studies on the role of social robots for language learning with the common variables of age groups, aspects of language, and affective gains. The collected samples in this study further used different forms of assessment (e.g., teacher-made and standard) and types of posttests (e.g., immediate and delayed). Therefore, a total of 12 common features were determined as potential moderator variables, namely, (a) name of robot, (b) age groups, (c) language proficiency, (d) design, (e) language domain, (f) type of assessment, (g) time point of assessment, (h) publication type, (i) educational level, (j) place, (k) setting, and (l) non-RALL condition type (see Table 1).

To investigate the extent to which these variables affect the effectiveness of RALL, the values for each were categorized into codes that could be compared both theoretically and statistically. The suggestion of Borenstein et al. (2009, 2022) to include the categories that have a minimum of eight effect sizes within each moderator was applied. For the variable of robots, three categories were identified: NAO, EngKey, and other. Randall (2019) reviewed studies on RALL and introduced NAO as the most widely used humanoid robot, which is autonomous and programmable. EngKey, however, is a teleoperated robot that is cartoon-like and egg-shaped. Apart from these two robots, which have key differences in both shape and system, the other robots were classified as *other*. Randall reported that younger students interacted with robots more than older students and were more interested.

TABLE 1

Summary of the included primary studies

No.	Name of the study	Robots	Age	Proficiency	Design	Language domain	Assessment	Publication type	Edu-level	Place	Setting	Non-RALL condition type
1	Ahmad et al. (2019)	NAO	7–12	Elementary	Exp	Vocabulary	Researcher's assessment	Journal	Elementary school	Other	ESL	Other technology
2	Alemi, M., & Haeri, N. (2020).	NAO	2–6	Elementary	Exp	Other	Researcher's assessment	Journal	Elementary school	Other	EFL	Traditional learning
3	Alemi et al. (2014)	NAO	7–12	Elementary	Exp	Vocabulary	Standard test	Journal	University	Other	EFL	Traditional learning
4	Balkibekov et al. (2016)	NAO	7–12	Elementary	Exp	Vocabulary	Researcher's assessment	Conference	Elementary school	Other	EFL	Other technology
5	De Wit et al. (2018)	NAO	2–6	Elementary	Quasi	Vocabulary	Researcher's assessment	Conference paper	Elementary school	Netherlands	EFL	Other technology
6	Hong, Z.-W., Huang, Y., Hsu, M., & Shen, W.-W. (2016)	Other	7–12	Elementary	Exp	Other	Test and questionnaire	Journal	Elementary school	Taiwan	EFL	Other technology
7	Hung, I. C., Chao, K.-J., Lee, L., & Chen, N.-S. (2013)	Other	Over 12	Intermediate	Not reported	Other	Researcher's assessment	Journal	University	Taiwan	EFL	Other technology
8	In, J., & Han, J. (2015)	Other	7–12	Elementary	Quasi	Other	Researcher's assessment	Conference paper	Elementary school	South Korea	EFL	Other technology
9	Kennedy, J., Baxter, P., Senft, E., & Belpaeme, T. (2016)	NAO	2–6	Elementary	Not reported	Vocabulary	Researcher's assessment	Conference paper	Elementary school	Other	ESL	Traditional learning
10	Kory Westlund, et al. (2017)	Other	2–6	Elementary	Not reported	Vocabulary	Researcher's assessment	Journal	Elementary school	Other	ESL	Traditional learning
11	Kory-Westlund, J. M., & Breazeal, C. (2019)	Other	2–6	Elementary	Quasi	Vocabulary	Researcher's assessment	Journal	Elementary school	Other	ESL	Other technology
12	Lee, S., Kim, C., Lee, J., Noh, H., Lee, K., & Lee, G. G. (2010)	EngKey	7–12	Elementary	Quasi	Affective	Test and questionnaire	Conference paper	Elementary school	South Korea	EFL	Traditional learning
13	Lee, S., Noh, H., Lee, J., Lee, K., Lee, G., Sagong, S., & Kim, M. (2011)	EngKey	7–12	Intermediate	Quasi	Other	Researcher's assessment	Journal	Elementary school	South Korea	EFL	Traditional learning

(continued)

TABLE 1. (CONTINUED)

No.	Name of the study	Robots	Age	Proficiency	Design	Language domain	Assessment	Publication type	Edu-level	Place	Setting	Non-RALL condition type
14	Leeuwestein et al. (2020)	NAO	2-6	Elementary	Exp	Vocabulary/Affective	Researcher's assessment	Journal	Elementary school	Netherlands	EFL	Other technology
15	Lio et al. (2019)	Other	Over 12	Elementary	Exp	Other	Researcher's assessment	Journal	University	Other	EFL	Other technology
16	Mazzoni, E., & Benvenuti, M. (2016)	Other	2-6	Elementary	Not reported	Vocabulary	Researcher's assessment	Journal	Elementary school	Other	EFL	Traditional learning
17	Kwon, O.-H., Koo, S.-Y., Kim, Y.-G., & Kwon, D.-S. (2010)	Other	7-12	Elementary	Quasi	Affective	Researcher's assessment	Conference paper	Elementary school	South Korea	EFL	Traditional learning
18	Park, S. J., Han, J. H., Kang, B. H., & Shin, K. C. (2011)	Other	Over 12	Elementary	Quasi	Vocabulary	Researcher's assessment	Conference paper	Elementary school	South Korea	EFL	Traditional learning
19	Rintjema, et al. (2018)	NAO	2-6	Elementary	Not reported	Vocabulary	Researcher's assessment	Conference paper	Elementary school	Netherlands	EFL	Traditional learning
20	Saerbeck, M., Schut, T., Barneck, C., & Janse, M. (2010)	Other	7-12	Elementary	Exp	Other	Researcher's assessment	Conference paper	Elementary school	Netherlands	EFL	Traditional learning
21	Schodde, T., Hoffmann, L., Stange, S., & Kopp, S. (2019)	NAO	2-6	Elementary	Exp	Affective	Test and questionnaire	Journal	Elementary school	Other	EFL	Other technology
22	Shin, J., & Shin, D. (2015)	NAO	Over 12	Intermediate	Quasi	Affective	Test and questionnaire	Conference paper	Elementary school	South Korea	EFL	Other technology
23	van den Berghe et al. (2020)	NAO	2-6	Elementary	Exp	Vocabulary	Researcher's assessment	Journal	Elementary school	Netherlands	EFL	Traditional learning
24	Van Minkelen et al. (2020)	NAO	2-6	Elementary	Quasi	Vocabulary	Researcher's assessment	Conference paper	Elementary school	Other	EFL	Other technology
25	Vogt et al. (2019)	NAO	2-6	Intermediate	Exp	Vocabulary	Researcher's assessment	Conference paper	Elementary school	Netherlands	EFL	Other technology
26	Westlund et al. (2015)	Other	2-6	Elementary	Not reported	Other	Researcher's assessment	Conference paper	Elementary school	South Korea	EFL	Other technology
27	Wu, W.-C. V., Wang, R.-J., & Chen, N.-S. (2015)	Other	Over 12	Intermediate	Not reported	Other	Researcher's assessment	Journal	University	Taiwan	EFL	Traditional learning

Note. Quasi = quasi-experimental, Exp = true-experimental.

Hence, for the age variable, three groups of 2 to 6 (preschoolers), 7 to 12 (elementary school students), and above 12 (older students) were identified. For the language proficiency variable, two values of *elementary* and *intermediate* were determined. Beginners have been the primary participants in RALL studies as this technology has not developed to the extent that it can benefit higher levels of language ability (Randall, 2019). In this meta-analysis, only 5 independent studies (out of 27) targeted the intermediate level (see Table 1).

For the design variable, the three codes of *experimental*, *quasi-experimental*, and *not reported* were identified. RALL studies have used various research designs (van den Berghe et al., 2019), but the most common is experimental, either true or quasi, depending on whether random assignment of the participants to control and experimental groups is employed or not. True experimental, which uses this randomization, is assumed to yield more accurate comparisons. Since most of the studies examined the effects of robots on learning new words (there were not enough studies addressing other skills to obtain the necessary sample size), the language domain was classified into *vocabulary* and *other*. The tests that were used to examine the effectiveness of RALL were classified into *teacher-made* and *standard* since validity and other psychometric properties can potentially affect the participants' performance (Bachman, 1990; Fulcher & Davidson, 2007). Additionally, the time point of assessment was coded to include *immediate* and *delayed* posttests. The time interval between the treatment and the posttest may influence the degree of RALL effects, so this variable was also examined. As for the variable of publication type, two groups of journal and conference papers were identified with the justification that studies published in journals may report higher effect sizes (Dickersin, 2005; Torgerson, 2006).

Although most RALL studies have been conducted at the elementary school level (Randall, 2019), a few were done at the university level; therefore, two values of *elementary* and *university* were coded for the variable of instructional level. Randall (2019) reported that almost 40% of the studies on RALL have taken place in Asia, for example, South Korea and Taiwan. While there is no justifiable reason for the result of RALL to be country-specific, Randall (2019, p. 16) argues that "due to cultural norms and different depictions of robots in the media," the effects of robots could vary as a result of the desired form and optimal nonverbal cues. Consequently, the variable of place was recognized as a potential moderator variable and classified into the groups of South Korea, Taiwan, Netherlands, and other. The variable of setting was divided into the two main and widely used values of English as a second language (ESL) and English as a foreign language (EFL). The former refers to contexts in which the students learn the target language in English-speaking countries, and the latter refers to the settings where the L2 learners acquire English within their native country or in places where the native speakers do not speak English. It is clear that ESL students have advantages over EFL learners as they have the opportunity to use English outside the classroom context with native speakers. This higher exposure to the target language can benefit L2 learners (Nation, 2006). The type of the control and/or comparison group was also examined; therefore, for the last variable, the non-RALL condition type was classified into two groups: traditional learning (i.e., without any technology) and other technology.

TABLE 2

Duval and Tweedie's trim and fill test of publication bias estimation

	Studies trimmed	Point estimate	Fixed effects	Point estimate	Random effects	<i>Q</i> value
			(Lower, upper)		(Lower, upper)	
Observed values		0.61	(0.54, 0.68)	0.68	(0.54, 0.78)	176.53
Adjusted values	11	0.53	(0.47, 0.60)	0.53	(0.40, 0.66)	248.98

Calculation of Effect Sizes and Meta-Analytic Procedures

The effect size calculations and meta-analytic procedures were conducted using Comprehensive Meta-Analysis software (CMA, version 4; ©2022, Biostat, Inc., Englewood, NJ). In this study, we used a standard mean difference for the main effect sizes, a weighted estimate of the difference between treatment and comparison groups (Lakens, 2013). We calculated the effect sizes as Hedges' *g*, which is the effect size measure that indicates the standardized difference between means and is considered less biased than Cohen's *d*, according to Kline (2004). Hedges' *g* is further suitable for small sample sizes, and the sample size of this study was relatively small. The positive effect size in this study suggests that language learners in a particular setting (i.e., experimental groups) outperformed language learners in another context (i.e., control groups). For the calculation of effect sizes, we employed adjusted posttest means that accounted for potential pretreatment differences. Using CMA, the researchers also calculated effect sizes based on a pretest-posttest control group design. For studies that reported using multiple measurement instruments for a certain variable, such as multiple choice and cloze tests, the average score from the variable was calculated (Borenstein et al., 2013). We additionally conducted a sensitivity analysis using the "one-study-removed" method to identify potential outliers of the pooled overall effects.

Publication Bias

To avoid or decrease publication bias in meta-analyses, we used a number of different procedures. First, trim and fill, a method developed by Duval and Tweedie (2000), was utilized to assess publication bias. The method uses a procedure to eliminate the most extreme small studies from the positive side of the funnel plot, iteratively recomputing the effect size until the funnel plot is symmetric (Borenstein et al., 2009). Under the fixed-effect model, the point estimate and 95% confidence interval for the combined studies is 0.61 (0.54, 0.68). Using trim and fill, these values are unchanged. Under the random-effects model, the point estimate and 95% confidence interval for the combined studies is 0.53 (0.40, 0.66). The results (Table 2 and Figure 2) indicate that there is no extreme value on the positive side that should be compensated for. In other words, our results were minimally affected by the publication bias.

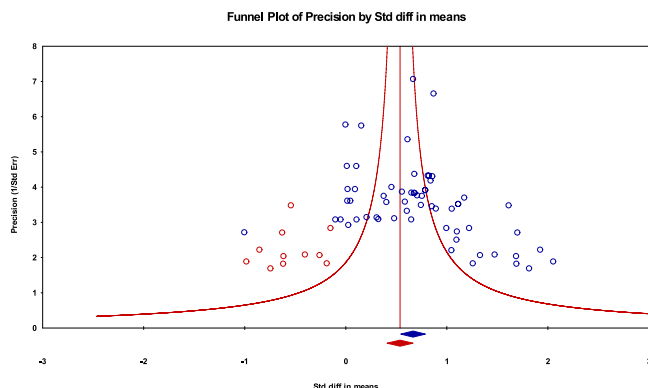


FIGURE 2. *Funnel plot of precision by effect sizes for the observed and imputed studies.*

Second, we added publication type as a moderator variable to investigate the possibility that published journals generated significantly larger effect sizes than unpublished papers. Third, Orwin's fail-safe N was high ($Nfs = 4,970$) (Orwin, 1983), indicating that we can be confident in this finding, and publication bias was negligible.

Missing Data

During the coding process, the researchers detected examples of missing data in six studies (see the appendix for the list of studies). We then contacted the corresponding author through email to obtain the missing data to compute the effect sizes. In some other studies, the researchers tried to calculate the effect size via raw data. Finally, one more effect size was added to our data pool.

Results

Overall Meta-Analysis Results

Out of 27 primary studies from 2010 to 2021, with 2,637 participants, 64 effect sizes (Hedges' g) were collected for meta-analysis. Table 3 and the forest plot in Figure 3 report the results of the average weighted Hedges' g , the 95% prediction intervals, the Q-test for heterogeneity, the two-tailed test of null, the between-study variance, and the percentage of variation between studies due to heterogeneity rather than sampling error. The effect sizes vary considerably from -0.92 to 10.48 . The overall effect size for the linguistic gain was found to be 0.78 , with a standard error of 0.10 , a z -value for a test of the null of 8.12 , and a corresponding p -value of less than $.001$. For the affective gain, the overall effect size was found to be 0.69 , with a standard error of 0.14 , a z -value for a test of the null of 4.80 , and a corresponding p -value of less than $.001$. Our grand mean for the combined linguistic and affective effect size of RALL on language learning achievement was found to be 0.75 , with a standard error of 0.08 , a z -value for a test of the null of 9.42 , and a corresponding p -value of less than $.001$. Based on Plonsky and Oswald (2014), the field-specific benchmarks for the effect size interpretation for L2 researchers, our average effect size was considered medium.

TABLE 3

Results of the univariate random-effects meta-analyses of RALL on language learning linguistic and affective achievement

Dependent variable	<i>N</i>	<i>k</i>	<i>g</i>	<i>SE</i>	95% CI	Test of null		Heterogeneity			Tau-Squared			
						<i>Z</i>	<i>P</i>	<i>Q</i>	<i>df</i>	<i>p</i>	<i>I</i> ²	τ^2	<i>SE</i>	τ
Linguistic gain	1,790	45	.78	.10	[.59, .96]	8.12	.00	271.64	44	.00	83.80	.31	.10	.55
Affective gain	847	19	.69	.14	[.41, .97]	4.80	.00	47.45	18	.00	62.06	.31	.14	.55
Overall gain	2,637	64	.75	.08	[.59, .90]	9.42	.00	319.20	63	.00	80.26	.30	.08	.55

Note. N = total number of participants, k = number of effect sizes, g = mean weighted effect size in Hedges' g, SE = standard error, CI = confidence interval, Z = Z value, p = p value, Q = Cochran's heterogeneity test; df = degrees of freedom, τ² = between-study variance; I² = percentage of variation between studies that is due to heterogeneity rather than sampling error.

The test of heterogeneity was significant, demonstrating that there was substantial variability between the studies ($Q = 319.20$, $df = 63$, $p < .001$, $I^2 = 80.26$). This also indicates that all the variance is unlikely to be due to sampling error, and we also conclude that the true effect size is likely to vary across studies. Therefore, we used the random effect model (Borenstein et al., 2013). Moreover, the significant Q and strong I^2 suggested that the moderator(s) might affect the results. Since this may be considered counter to the heterogeneity of the results, considering that both Q and I^2 were indicators of heterogeneity, we decided to run a moderator analysis as well (see Tables 4 to 6). For the sensitivity analysis, one-study-removed analyses were conducted via CMA. The results showed that if one study were left out, the overall effect would not change substantially ($g = 0.75$, $p < .001$, 95% CI [0.59, .90], $I^2 = 80.26\%$) (Viechtbauer & Cheung, 2010).

Moderator Analyses

To answer the second research question, the moderating factors of RALL on language learning achievement were investigated. Independent meta-regression analyses for each group of moderator variables were conducted through sets of characteristics. Q -tests were used to evaluate if a particular variable was a significant moderator. The moderating effect of each set, as a linked group of moderators, was also assessed to examine the effect of the whole set. We scrutinized three main sets of variables, which included design characteristics, participants' characteristics, and language characteristics. Detailed information is presented in Tables 4 to 6.

Design Characteristics

In our design set, as shown in Table 4, the results of meta-regression analyses for the type of publication, journal article, and conference journal ($Q = 0.07$, $df = 1$, $p = .78$, $\tau^2 = .30$, $I^2 = 80.46$, $R^2 = .00$), for the design of the study, experimental, quasi-experimental and not reported ($Q = 3.57$, $df = 2$, $p = .16$, $\tau^2 = .30$, $I^2 = 80.46$, $R^2 = .02$), for the experiment time points, first and delay ($Q = 0.59$, $df = 1$, $p = .44$, $\tau^2 = .30$, $I^2 = 80.23$, $R^2 = -.01$), for the study's setting, EFL and ESL

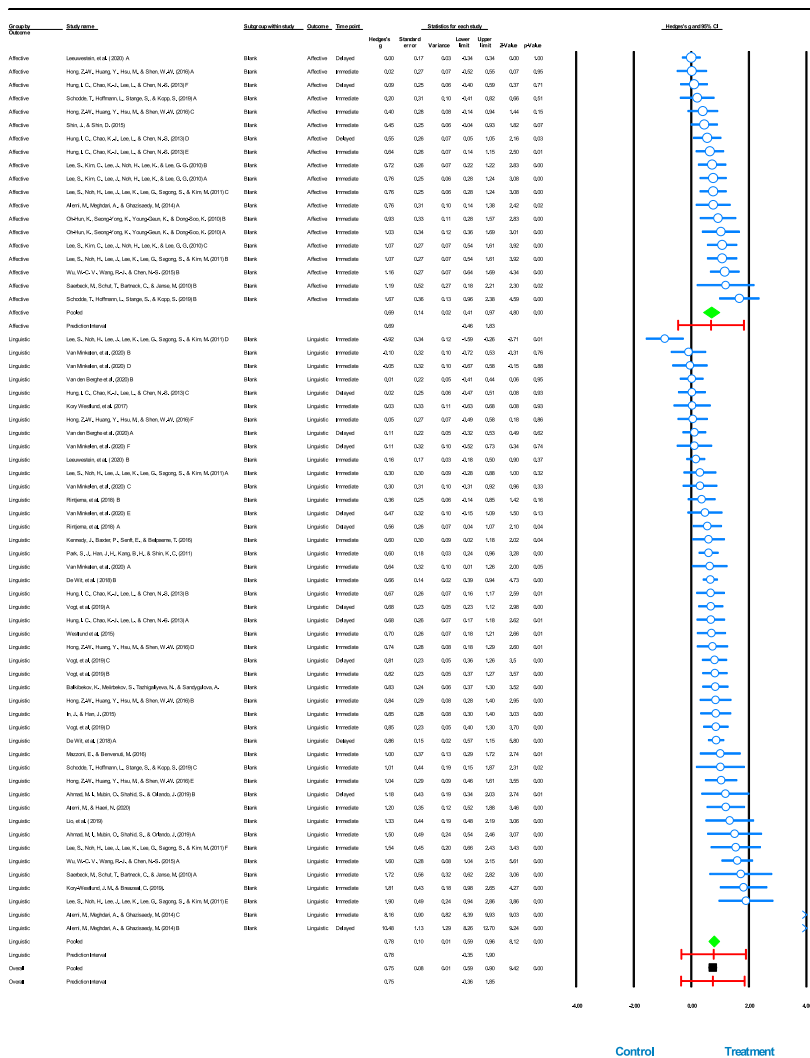


FIGURE 3. Forest plot of included effect and prediction interval.

($Q = 0.45$, $df = 1$, $p = .50$, $\tau^2 = .30$, $I^2 = 80.34$, $R^2 = .00$), and for between-group and within-group comparison ($Q = 0.72$, $df = 1$, $p = .39$, $\tau^2 = .29$, $I^2 = 79.96$, $R^2 = .01$), showed that they had no significant influence on the effects of RALL on language achievement. The results for the place of the study, the Netherlands,

TABLE 4
Moderator analysis of design characteristics

Moderator	<i>N</i>	<i>k</i>	<i>g</i>	95% CI	<i>Q</i> ^b	<i>df</i>	<i>p</i> ^b	τ^2	<i>I</i> ²	<i>R</i> ²
Publication type					0.07	1	.78	.30	80.46	.00
Journal article	1810	46	0.76	[0.58, 0.95]						
Conference journal	827	18	0.72	[0.43, 1.00]						
Place of the study					3.06	3	.38	.14	66.54	.00
The Netherlands	947	18	0.52	[0.30, 0.74]						
South Korea	280	14	0.74	[0.49, 1.00]						
Taiwan	812	15	0.78	[0.52, 1.01]						
Others	598	14	0.60	[0.36, 0.84]						
Design of the study					3.57	2	.16	.30	80.46	.02
Experimental	1,298	27	0.90	[0.65, 1.14]						
Quasi-experimental	833	26	0.72	[0.48, 0.96]						
Not reported	506	11	0.48	[0.11, 0.84]						
Time points					0.59	1	.44	.30	80.23	-.01
Immediate	1,785	50	0.78	[0.60, 0.96]						
Delayed	852	14	0.63	[0.31, 0.96]						
Name of the robots					0.00	2	.99	.30	80.67	.00
EngKey	140	9	0.76	[0.34, 1.18]						
NAO	1,452	30	0.75	[0.52, 0.97]						
Others	1,045	25	0.75	[0.50, 1.00]						
Setting					0.45	1	.50	.30	80.34	.00
EFL	2,166	55	0.73	[0.56, 0.89]						
ESL	471	9	0.88	[0.46, 1.30]						
Non-RALL condition type					0.63	1	.42	.13	65.78	.00
Traditional learning	721	21	0.72	[0.51, 0.93]						
Other technology	1,824	40	0.61	[0.47, 0.76]						
Design characteristics set					27.71	13	.00	.31	84.00	.00

Note. *N* = total number of participants, *k* = number of effect sizes, *g* = mean weighted effect size in Hedges' *g*, CI = confidence interval, *p* = *p*-value, *Q*^b = *Q*-between, *df* = degrees of freedom, τ^2 = between-study variance; *I*² = percentage of variation between studies that is due to heterogeneity rather than sampling error. *R*² = the proportion of the original variance explained by the covariates.

South Korea, Taiwan, and others) (*Q* = 3.06, *df* = 3, *p* = .38, τ^2 = .14, *I*² = 66.54., *R*² = .00) indicated that these moderators had no significant effects on the influence of RALL on language achievement. In other words, RALL was equally effective in all these places. Finally, for the robots, EngKey, NAO, and others (*Q* = 0.0, *df* = 2, *p* = .99, τ^2 = .30, *I*² = 80.67, *R*² = .0), and non-RALL condition type, traditional learning condition and other technologies (*Q* = 0.63, *df* = 1, *p* = .42, τ^2 = .13, *I*² = 65.78, *R*² = .0), no significant differences were found. Robots were all effective in their corresponding settings. The findings for the design set, as a linked group of all the moderators (*Q* = 27.71, *df* = 13, *p* = .00, τ^2 = .31, *I*² = 80.00, *R*² = .00) indicated a significant effect of RALL on language learning achievement, whereas the effects of design moderators at the individual level, not as a set, were not significant.

TABLE 5
Moderator analysis for participants' characteristics

Moderator	<i>N</i>	<i>k</i>	<i>g</i>	95% CI	<i>Q</i> ^b	<i>df</i>	<i>p</i> ^b	τ^2	<i>I</i> ²	<i>R</i> ²
Participant's age					9.40	2	.00	.28	79.50	.04
2-6	1,274	27	0.55	[0.32, 0.78]						
7-12	882	28	1.03	[0.79, 1.27]						
Over 12	481	9	0.54	[0.15, 0.93]						
L2 proficiency level					1.29	1	.25	.30	80.53	.00
Elementary	1,824	50	0.80	[0.62, 0.97]						
Intermediate	813	14	0.58	[0.26, 0.91]						
L2 proficiency measurement					1.01	1	.31	.29	80.28	.00
Researcher's assessment	1,903	46	0.70	[0.52, 0.88]						
Test and questionnaire	734	18	0.87	[0.58, 1.17]						
Instructional level					0.78	2	.37	.13	65.61	.00
Elementary school	2,130	53	0.67	[0.54, 0.79]						
University level	447	8	0.51	[0.20, 0.83]						
Participants characteristics set					5.73	5	.00	.33	64.99	.00

Note. *N* = total number of participants, *k* = number of effect sizes, *g* = mean weighted effect size in Hedges' *g*, *CI* = confidence interval, *p* = *p* value, *Q*^b = *Q*-between, *df* = degrees of freedom, τ^2 = between-study variance; *I*² = percentage of variation between studies that is due to heterogeneity rather than sampling error. *R*² = the proportion of the amount of heterogeneity accounted for.

Participants Characteristics

The next set of moderators was the participants' characteristics, as depicted in Table 5. The results of meta-regression analyses for the participants' L2 proficiency levels, including elementary and intermediate ($Q = 1.29$, $df = 1$, $p = .25$, $\tau^2 = .30$, $I^2 = 80.53$, $R^2 = .00$), for the L2 proficiency measurement scales, including researcher's assessment and test/questionnaire ($Q = 1.01$, $df = 1$, $p = .31$, $\tau^2 = .29$, $I^2 = 80.28$, $R^2 = .00$), and the participants' instructional level ($Q = 0.78$, $df = 1$, $p = .37$, $\tau^2 = .13$, $I^2 = 65.61$, $R^2 = .00$) showed no significant influence on the effects of RALL on language achievement. For the moderating effect of the participants' age ($Q = 9.40$, $df = 2$, $p = .00$, $\tau^2 = .28$, $I^2 = 79.50$, $R^2 = .04$), the results indicated a significant impact for this moderator. Language learners between the ages of 7 to 12 gained significantly higher scores than the other age groups. The participants' age accounted for 4% of the amount of heterogeneity, which is substantial. The findings for the participants' set, as a linked group of all the nine moderators ($Q = 5.73$, $df = 5$, $p = .33$, $\tau^2 = .33$, $I^2 = 64.99$, $R^2 = .00$), indicated no significant effect.

Language Characteristics

For the last set of moderators (Table 6), we calculated the effect of RALL on two groups of language skills and components, namely vocabulary learning and other language abilities ($Q = 0.26$, $df = 1$, $p = .60$, $\tau^2 = .39$, $I^2 = 84.15$, $R^2 = .00$). Studies that did not focus on vocabulary were not independently large enough to be included in a separate group. They studied reading, speaking,

TABLE 6
Moderator analysis for language characteristics

Moderator	<i>N</i>	<i>k</i>	<i>g</i>	95% CI	<i>Q</i> ^b	<i>df</i>	<i>p</i> ^b	τ^2	<i>I</i> ²	<i>R</i> ²
					0.26	1	.60	.39	84.15	.00
Other	479	19	.74	[0.42, 1.05]						
Vocabulary	1,311	26	.85	[0.57, 1.12]						

Note. *N* = total number of participants, *k* = number of effect sizes, *g* = mean weighted effect size in Hedges' *g*, *CI* = confidence interval, *p* = *p* value, *Q*^b = *Q*-between, *df* = degrees of freedom, τ^2 = between-study variance; *I*² = percentage of variation between studies that is due to heterogeneity rather than sampling error. *R*² = the proportion of the amount of heterogeneity accounted for.

writing, listening, pronunciation, and pragmatics. The highest frequency among them belonged to reading (*k* = 6) and the lowest to pragmatics (*k* = 1). So, we decided to put them collectively in one group to run the moderator analysis. The results showed no significant differences concerning the effect of RALL on the two groups. For other language skills and components, such as listening, speaking, etc., moderator analyses were not regarded as meaningful, as the total number of studies was insufficient; therefore, most moderating variables could not be meaningfully compared.

Discussion

This meta-analysis explored the overall effects of RALL and the moderators that affect the effect size. Initially, the results of Duval and Tweedie's publication bias test showed no extreme values in the included effect sizes. Therefore, the results can be considered minimally influenced by publication bias. The overall effect size was calculated using Hedges' *g* through the random effect model as the variation of the effect sizes proved heterogeneous. The overall effect size was *g* = 0.78, representing a large effect size. In other words, the included samples showed an average of 0.78 of the pooled standard deviation differences between the post-test scores. The values were close for both linguistic (*g* = 0.69) and affective gains (*g* = 0.75).

The Linguistic and Affective Gains of RALL

With regards to the linguistic gains, two types of dependent variables were included in the studies: vocabulary learning (*N* = 26) and other language abilities (*N* = 19), both of which showed relatively close *g* values of 0.85 and 0.74. This may be attributed to the limited abilities of robots. Although robots have certain advantages like providing an authentic interaction environment (van den Berghe et al., 2019), adapting nonverbal actions (e.g., Kanero et al., 2018; de Wit et al., 2018), and providing individual training (Aleami & Bahramipour, 2019; Demir-Lira et al., 2020; Mazzoni & Benvenuti, 2015; Schodde et al., 2019), their limited programming capacity may be the reason that they are primarily used to enhance linguistic features, which may limit their scope. They are often used to teach pre-programmed vocabulary, reading ability of a certain set of texts, certain speaking

patterns, grammatical structures, or pronunciation of certain words, as these dimensions of language are more easily programmed and tested in robotic studies. However, pragmatics, as a less concrete aspect, cannot yet be investigated in RALL. Nevertheless, the moderator analysis showed that the cumulative effect sizes in the two groups are similarly large ($g = 0.83$ for both groups).

As Randall (2019) explains, the positive effects of RALL for linguistic gains align with the interactionist theory of language learning, which is linked to the utilization of robots. The interactionist theory proposes that the most effective way of acquiring a language is through social communication with others. This interaction should be tailored to an individual's zone of proximal development (ZPD), which refers to what they could achieve with assistance (Nor & Ab Rashid, 2018). Employing a robot as an embodied agent that is easily regarded as a social entity capable of communication may make it more effective than other forms of technology in facilitating language acquisition. The positive findings of RALL for language learning could be attributed to some of the unique features of robots, such as their physical presence (H. Lee & Lee, 2021; Li, 2015; Randall, 2019; van den Berghe et al., 2019) and interactivity, which is facilitated by their social behaviors.

Regarding the affective factors, the cumulative effect size was also large ($g = 0.69$). Robots make the learning environment interesting (H. L. Chen et al., 2020; Freyer et al., 2019) through their autonomous or semiautonomous behavior and/or provision of a virtual communicative setting. Reducing anxiety and enhancing motivation are affective factors that are frequently addressed in the literature (e.g., Ahmad et al., 2019; Alemi et al., 2014b; de Haas et al., 2020; Hong et al., 2016; Hung et al., 2013; S. Lee et al., 2011). The clearest benefit of using robots over other forms of technology is that they lower the affective filter. Drawing from Krashen's affective filter theory and studies highlighting the impact of emotions on language acquisition (Sheen, 2008), it is likely that robots have the potential to reduce the affective barrier and enhance the learners' ability to understand input, ultimately facilitating the acquisition of language skills. Consistent with the results of this meta-analysis, EFL participants in Alemi et al. (2017) exhibited reduced anxiety levels, elevated enthusiasm levels, and greater involvement following repeated interactions with a robot.

In a review study, Randall (2019) reported positive effects of RALL on affective factors like motivation, interest, engagement, confidence, and anxiety. Robots' verbal and nonverbal immediacy was attributed to the robots' contributions to these factors. Also, the sense of achievement and relaxed atmosphere created by robots have been identified as a justification for the positive role of RALL in building confidence and willingness to communicate (Hong et al., 2016). van den Berghe et al. (2019) further conducted a review study and reported significant associations between RALL and language-related emotions like motivation, confidence, and engagement, whereas other forms of technology, such as interactive whiteboards and VR, have reported weaker evidence of positive effects on affective variables (Golonka et al., 2014). Furthermore, linguistic achievements can result in affective gains since the two variables are interrelated (Pekrun et al., 2002).

Suggestions for RALL Based on the Moderator Analyses

In this study, we identified 12 potential moderator variables to probe for their effect on RALL's effectiveness. Triggered by the substantial variability in the range of obtained effect sizes, we conducted moderator analysis and found that the variable of language domain showed no significant moderating effect. Other factors that showed no significant moderating effects were the type of publication (journal or conference), design (experimental, quasi-experimental, or not reported), the time point of assessment (immediate or delayed), setting (EFL or ESL), type of robots (EngKey, NAO, others), place of study (Netherlands, South Korea, Taiwan, others), L2 proficiency (elementary or intermediate), type of assessment (researcher's assessment or test/questionnaire), instructional level (elementary school and university), and non-RALL condition (traditional learning and other technology). These results indicate that the effectiveness of robot-assisted language learning is irrespective of extraneous variables like the learner's language background and potential measurement errors of the instruments used in the studies.

As for the design-related variables, Dickersin (2005) hypothesized that studies published in peer-reviewed journals usually report larger effect sizes than unpublished ones or studies presented at conferences; however, this was not found in this study since RALL is a relatively new area of study, with high-profile conferences held to discuss related issues. Nevertheless, more robust empirical studies need to be conducted to find whether effect sizes vary across studies published in journals and those presented at conferences (Zhang & Zhang, 2022). The place of the study did not act as a moderator variable either, which may be due to the cultural similarities between the Asian countries of South Korea and Taiwan, two of the studied categories. However, the inclusion of the Netherlands may tentatively suggest that RALL could be effective regardless of place, but further investigation of various cultures is needed.

The random assignment of the participants into control and experimental groups did not play a significant role in the effectiveness of RALL. This factor was not considered in previous review studies, suggesting that differences in effect sizes do not arise from randomization. The same is true for the time point of assessment, as the administration of immediate and delayed posttests did not affect the degree of RALL effectiveness. Therefore, future RALL studies may choose to implement either form of experimental design (true or quasi) and post-test (immediate or delayed). The results for the type of robots used also indicated that EngKey and NAO, as teleoperated and autonomous robots, respectively, did not show significantly different effect sizes, indicating that both functions can be used in RALL without any significant changes in effectiveness. Randall (2019) further argued that both autonomous and teleoperated robots are used in RALL, and the former type is even sometimes remotely controlled in some experiments. The EFL and ESL division was also insignificant in RALL effectiveness, probably because more exposure to the target language (as in the ESL setting) benefits more proficient students and productive skills, while RALL is mainly used for elementary levels and to teach basic words (van den Bergh et al., 2019). Similarly, no statistically significant difference was observed for the variable of non-RALL condition type. This finding indicates that RALL is effective regardless of the instruction used in the control and/or comparison group.

As for the participant-related factors, age was the only moderator variable in determining the effectiveness of RALL. The age group of 7 to 12 showed a significantly higher effect size compared to those above 12 and those in the age range of 2 to 6. Fernández-Llamas et al. (2018) stated that younger learners exhibited greater familiarity with robots. Multiple factors may contribute to this phenomenon, one of which can be attributed to children's natural aptitude for language acquisition compared to adults (Singleton & Leśniewska, 2021). Moreover, first-grade pupils display greater interest levels and prolonged interaction with the robot as opposed to sixth-graders (Kanda et al., 2004). Therefore, it is recommended that language educators employ social robots for younger learners, though with caution, as RALL has its own shortcomings, as explained in the introduction of this article. The language proficiency level of the participants did not show significant effect sizes, suggesting that while RALL can be used for the intermediate level as well, the current state of robots is suitable mainly for primary levels (Randall, 2019; van den Berghe et al., 2019) as more complex systems need to be developed for higher levels. The insignificant role of type of assessment in RALL's effectiveness also reveals that psychometric properties of tests are not critical in RALL research. Thus, posttests can be either researcher-made or standard. Similarly, the instructional level of the participants did not make a difference in the effect sizes. However, this finding will need further investigation after robots are developed to target the university level because most of the included samples addressed the elementary school level.

The classification of the language domain only included the vocabulary aspect, a well-documented role in RALL, with many studies targeting this domain. However, few studies have addressed other skills and subskills, suggesting that robots, in their current state, can be employed to teach new words at the elementary level. The variable of language domain was not identified as a moderator when the category of *vocabulary* is compared to *other*. This finding shows that the effectiveness of RALL is not different across aspects of the language, but due to a small sample size for the other dimensions of the language that were merged, further investigations are required in the future when the robots are suitable to target receptive and productive skills of the language.

Conclusion and Directions for Future Research

The current study probed the effectiveness of RALL and the extent to which this effectiveness can be influenced by a selected number of moderators through a meta-analysis. The results showed that robots are effective in improving both linguistic and affective gains. Their effectiveness is not limited to any specific participant or design factor. Although some moderating factors were at play, the moderators relating to these factors showed medium to very large effect sizes.

Despite the implications of this meta-analysis, a number of limitations could have influenced the reliability of the findings. The first limitation was the amount of missing data needed for secondary analysis in some of the targeted papers. As mentioned, we contacted the authors to provide us with the raw data, but only a few responded. Nevertheless, with a total sample size of 64 effect sizes from 2,637 participants, the results of this meta-analysis yielded sufficient statistical power, but future studies may attempt to obtain more samples with less missing

data. The next limitation was in the scope of synthesized studies. While there were studies targeting a different aspect of language, the majority of the studies focused on vocabulary learning. Consequently, we had to run the moderator analysis on two groups of dependent variables, that is, vocabulary and other language abilities. Hence, more research on other language components is required.

In addition to some suggestions for large-scale studies, more specific recommendations for individual empirical studies are evident. First, future research should employ qualitative or mixed-methods designs to explore the effectiveness of RALL in affective and cognitive factors. Second, given the potential of social robots to increase students' willingness to communicate and their capability to act like partners in conversations (C.-W. Chang et al., 2010), future studies may examine the effects of humanoid robots on language learners' speaking competence. Third, as the design of robots advances, future investigations should examine their effects on other skills and sub-skills. Finally, further experimental studies should investigate the effectiveness of robots versus other forms of technology to find if RALL is more influential than CALL or mobile-assisted language learning (MALL).

Appendix

List of Papers With Missing Data

Title	Corresponding Author
Robot-Assisted Language Learning: Artificial Intelligence in Second Language Acquisition	Dara Tafazoli
Improvement of Japanese Adults' English Speaking Skills via Experiences Speaking to a robot	Takamasa Iio
Paper-Learning Effect of Implicit Learning in Joining-in-Type Robot-Assisted Language Learning System	AlBara Khalifa
Sociable Robot Improves Toddler Vocabulary Skills	Javier R. Movellan
Investigating the Effects of a Robot Peer on L2 Word Learning	Rianne van den Berghe
A Comparison of Children Learning New Words From Robots, Tablets, & People	Jacqueline Kory Westlund

ORCID iDs

Ali Derakhshan  <https://orcid.org/0000-0002-6639-9339>

Esmaeel Saeedy Robat  <https://orcid.org/0000-0001-9899-8523>

Mostafa Janebi Enayat  <https://orcid.org/0000-0001-8724-9121>

Notes

This work was supported by Golestan University. The authors wish to thank Golestan University for its support. They are also grateful for the insightful comments suggested by the editor and the anonymous reviewers.

This study is supported by Golestan University under Grant number 1515.

References

References marked with an asterisk indicate studies included in the meta-analyses.

- *Ahmad, M. I., Mubin, O., Shahid, S., & Orlando, J. (2019). Robot's adaptive emotional feedback sustains children's social engagement and promotes their vocabulary learning: A long-term child robot interaction study. *Adaptive Behavior*, 27(4), 243–266. <https://doi.org/10.1177/1059712319844182>
- Alemi, M., & Bahramipour, S. (2019). An innovative approach of incorporating a humanoid robot into teaching EFL learners with intellectual disabilities. *Asian-Pacific Journal of Second and Foreign Language Education*, 4(1). <https://doi.org/10.1186/s40862-019-0075-5>
- *Alemi, M., & Haeri, N. (2020). Robot-assisted instruction of L2 pragmatics: Effects on young EFL learners' speech act performance. *Language Learning & Technology*, 24(2), 86–103.
- *Alemi, M., Meghdari, A., & Ghazisaedy, M. (2014a). Employing humanoid robots for teaching English language in Iranian junior high-schools. *International Journal of Humanoid Robotics*, 11(03), 1450022. <https://doi.org/10.1142/s0219843614500224>
- Alemi, M., Meghdari, A., & Ghazisaedy, M. (2014b). The effect of employing humanoid robots for teaching English on students' anxiety and attitude. In *Second RSI/ISM International Conference on Robotics and Mechatronics (ICRoM)* (pp. 754–759). <https://doi.org/10.1109/ICRoM.2014.6990994>
- Alemi, M., Meghdari, A., & Ghazisaedy, M. (2015). The impact of social robotics on L2 learners' anxiety and attitude in English vocabulary acquisition. *International Journal of Social Robotics*, 7, 523–535. <https://doi.org/10.1007/s12369-015-0286-y>
- Alemi, M., Meghdari, A., & Sadat Haeri, N. (2017). Young EFL learners' attitude towards RALL: An observational study focusing on motivation, anxiety, and interaction. In *Proceedings of the International Conference on Social Robotics 2017* (pp. 252–261). Springer International. <https://doi.org/10.1007/978-3-319-70022-9>
- Bachman, F. (1990). *Fundamental considerations in language testing*. Oxford University Press.
- *Balkibekov, K., Meiirbekov, S., Tazhigaliyeva, N., & Sandygulova, A. (2016). Should robots win or lose? Robot's losing playing strategy positively affects child learning. *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)* (pp. 706–711). IEEE. <https://doi.org/10.1109/ROMAN.2016.7745196>.
- Belpaeme, T., Kennedy, J., Ramachandran, A., Scassellati, B., & Tanaka, F. (2018). Social robots for education: A review. *Science Robotics*, 3(21). <https://doi.org/10.1126/scirobotics.aat5954>
- Bernard, R. M., Abrami, P. C., Lou, Y., Borokhovski, E., Wade, A., Wozney, L., Walset, P. A., Fiset, M., & Huang, B. (2004). How does distance education compare with classroom instruction? A meta-analysis of the empirical literature. *Review of Educational Research*, 74(3), 379–439.
- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to meta-analysis*. John Wiley & Sons, Ltd. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470743386.ch4>
- Borenstein, M., Hedges, L. E., Higgins, J. P. T., & Rothstein, H. R. (2022). *Comprehensive meta-analysis Version 4*. Biostat, Inc. www.Meta-Analysis.com
- Chang, C.-W., Lee, J.-H., Chao, P.-Y., Wang, C.-Y., & Chen, G.-D. (2010). Exploring the possibility of using humanoid robots as instructional tools for teaching a second

- language in primary school. *Journal of Educational Technology & Society*, 13(2), 13–24. <http://www.jstor.org/stable/jeductechsoci.13.2.13>
- Chen, H. J. H., & Yang, T. U. C. (2013). The impact of adventure video games on foreign language learning and the perceptions of learners. *Interactive Learning Environments*, 21(2), 129–141. <https://doi.org/10.1080/10494820.2012.705851>
- Chen, H. L., Widarso, G. V., & Sutrisno, H. (2020a). A chatbot for learning Chinese: Learning achievement and technology acceptance. *Journal of Educational Computing Research*, 58(6), 1161–1189. <https://doi.org/10.1177%2F0735633120929622>
- Chen, Y., Smith, T. J., York, C. S., & Mayall, H. J. (2020b). GoogleEarth virtual reality and expository writing for young English learners from a funds of knowledge perspective. *Computer Assisted Language Learning*, 33(1–2), 1–25. <https://doi.org/10.1080/09588221.2018.1544151>
- Dashtestani, R. (2016). Moving bravely towards mobile learning: Iranian students' use of mobile devices for learning English as a foreign language. *Computer Assisted Language Learning*, 29(4), 815–832. <https://doi.org/10.1080/09588221.2015.1069360>
- De Haas, M., Vogt, P., & Krahmer, E. (2020). The effects of feedback on children's engagement and learning outcomes in robot-assisted second language learning. *Frontiers in Robotics and AI*, 7, 1–18. <https://dx.doi.org/10.3389%2Ffront.2020.00101>
- *De Wit, J., Schodde, T., Willemsen, B., Bergmann, K., de Haas, M., Kopp, S., Krahmer, E., & Vogt, P. (2018). The effect of a robot's gestures and adaptive tutoring on children's acquisition of second language vocabularies. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (pp. 50–58). <https://doi.org/10.1145/3171221.3171277>
- Demir-Lira, Ö. E., Kanero, J., Oranç, C., Koskulu, S., Franko, I., Göksun, T., & Küntay, A. C. (2020). L2 vocabulary teaching by social robots: The role of gestures and on-screen cues as scaffolds. *Frontiers in Education*. Advance online publication. <https://doi.org/10.3389/feduc.2020.599636>
- Dickersin, K. (2005). Publication bias: Recognizing the problem, understanding its origin and scope, and preventing harm. In H. R. Rothstein, A. J. Sutton, & M. Borenstein (Eds.), *Publication bias in meta-analysis: Prevention, assessment and adjustments* (pp. 11–33). Wiley.
- Duval, S., & Tweedie, R. (2000). Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics*, 56(2), 455–463.
- Fernández-Llamas, C., Conde, M. A., Rodríguez-Lera, F. J., Rodríguez-Sedano, F. J., & García, F. (2018). May I teach you? Students' behavior when lectured by robotic vs. human teachers. *Computers in Human Behavior*, 80(2), 460–469. <https://doi.org/10.1016/j.chb.2017.09.028>
- Fryer, L. K., Nakao, K., & Thompson, A. (2019). Chatbot learning partners: Connecting learning experiences, interest and competence. *Computers in Human Behavior*, 93, 279–289. <https://doi.org/10.1016/j.chb.2018.12.023>
- Fulcher, G., & Davidson (2007). *Language testing and assessment: An advanced resource book*. Routledge Applied Linguistics.
- Golonka, E. M., Bowles, A. R., Frank, V. M., Richardson, D. L., & Freynik, S. F. (2014). Technologies for foreign language learning: A review of technology types and their effectiveness. *Computer Assisted Language Learning*, 27(1), 70–105. <https://doi.org/10.1080/09588221.2012.700315>

- *Hong, Z. W., Huang, Y. M., Hsu, M., & Shen, W. W. (2016). Authoring robot-assisted instructional materials for improving learning performance and motivation in EFL classrooms. *Educational Technology & Society*, 19 (1), 337–349.
- Huang, J. X., Lee, K. S, Kwon, O. W., & Kim, Y. K. (2017). A chatbot for a dialogue-based second language learning system. In K. Borthwick, L. Bradley, & S. Thouësny (Eds), *CALL in a climate of change: Adapting to turbulent global conditions – short papers from EUROCALL 2017* (pp. 151–156). <https://doi.org/10.14705/rpnet.2017.eurocall2017.705>
- *Hung, I. C., Chao, K.-J., Lee, L., & Chen, N.-S. (2013). Designing a robot teaching assistant for enhancing and sustaining learning motivation. *Interactive Learning Environments*, 21(2), 156–171. <https://doi.org/10.1080/10494820.2012.705855>
- Iio, T., Maeda, R., Ogawa, K., Yoshikawa, Y., Ishiguro, H., Suzuki, K., Aoki, T., Maesaki, M., & Hama, M. (2018). Improvement of Japanese adults' English speaking skills via experiences speaking to a robot. *Journal of Computer Assisted Language Learning*, 35(2), 228–245. <https://doi.org/10.1111/jcal.12325>
- *In, J.-Y., & Han, J.-H. (2015). The acoustic-phonetics change of English learners in robot assisted learning. In *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction* (pp. 39–40). <https://dx.doi.org/10.1145/2701973.2702003>
- Janebi Enayat, M., & Haghighatpasand, M. (2019). Exploiting adventure video games for second language vocabulary recall: A mixed-methods study. *Innovation in Language Learning and Teaching*, 13(1), 61–75. <https://doi.org/10.1080/17501229.2017.1359276>
- Jiang, L., & Yu, S. (2020). Appropriating automated feedback in L2 writing: Experiences of Chinese EFL student writers. *Computer Assisted Language Learning*. Advance online publication. <https://doi.org/10.1080/09588221.2020.1799824>
- Kanda, T., Hirano, T., Eaton, D., & Ishiguro, H. (2004). Interactive robots as social partners and peer tutors for children: A field trial. *Human-Computer Interaction*, 19(1–2), 61–84.
- Kanero, J., Geçkin, V., Oranç, C., Mamus, E., Küntay, A. C., & Göksun, T. (2018). Social robots for early language learning: Current evidence and future directions. *Child Development Perspectives*, 12(3), 146–151. <https://doi.org/10.1111/cdep.12277>
- *Kennedy, J., Baxter, P., Senft, E., & Belpaeme, T. (2016). Social robot tutoring for child second language learning. In *11th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 231–238). <https://doi.org/10.1109/HRI.2016.7451757>
- Kline, R. (2004). *Beyond significance testing: Reforming data analysis methods in behavioral research*. Amer Psychological Assn.
- *Kory-Westlund, J. M., & Breazeal, C. (2019). A long-term study of young children's rapport, social emulation, and language learning with a peer-like robot playmate in preschool. *Frontiers in Robotics and AI*, 6(81). <https://doi.org/10.3389/frobt.2019.00081>
- *Kory Westlund, J. M., Dickens, L., Jeong, S., Harris, P. L., DeSteno, D., & Breazeal, C. L. (2017). Children use non-verbal cues to learn new words from robots as well as people. *International Journal of Child-Computer Interaction*, 13, 1–9. <https://doi.org/10.1016/j.ijcci.2017.04.001>

- *Kwon, O.-H., Koo, S.-Y., Kim, Y.-G., & Kwon, D.-S. (2010). Telepresence robot system for English tutoring. In *IEEE Workshop on Advanced Robotics and Its Social Impacts*. <https://doi.org/10.1109/ARSO.2010.5679999>
- Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4(863). <https://doi.org/10.3389/fpsyg.2013.00863>
- Lee, H., & Lee, J. H. (2021). Social robots for English language teaching. *ELT Journal*. Advance online publication. <https://doi.org/10.1093/elt/ccab041>
- *Lee, S., Kim, C., Lee, J., Noh, H., Lee, K., & Lee, G. G. (2010). Affective effects of speech-enabled robots for language learning. In *IEEE Spoken Language Technology Workshop* (pp. 145–150). <https://doi.org/10.1109/SLT.2010.5700837>
- *Lee, S., Noh, H., Lee, J., Lee, K., Lee, G., Sagong, S., & Kim, M. (2011). On the effectiveness of robot-assisted language learning. *ReCALL*, 23(1), 25–58. <https://doi.org/10.1017/S0958344010000273>
- *Leeuwestein, H., Barking, M., Sodaci, H., Oudgenoeg-Paz, O., Verhagen, J., Vogt, P., Aarts, R., Spit, S., de Haas, M., de Wit, J., & Leseman, P. (2021). Teaching Turkish-Dutch kindergartners Dutch vocabulary with a social robot: Does the robot's use of Turkish translations benefit children's Dutch vocabulary learning? *Journal of Computer Assisted Learning*, 37(3), 603–620. <https://doi.org/https://doi.org/10.1111/jcal.12510>
- Li, J. (2015). The benefit of being physically present: A survey of experimental works comparing copresent robots, telepresent robots and virtual agents. *International Journal of Human-Computer Studies*, 77, 23–37. <https://doi.org/10.1016/j.ijhcs.2015.01.001>
- Lin, E. J., & Lin, H. (2019). Mobile-assisted ESL/EFL vocabulary learning: A systematic review and meta-analysis. *Computer Assisted Language Learning*, 32(8), 878–919. <https://doi.org/10.1080/09588221.2018.1541359>
- *Lio, T., Maeda, R., Ogawa, K., Yoshikawa, Y., Ishiguro, H., Suzuki, K., Aoki, T., Maesaki, M., & Hama, M. (2019). Improvement of Japanese adults' English speaking skills via experiences speaking to a robot. *Journal of Computer Assisted Learning*, 35(2), 228–245. <https://doi.org/https://doi.org/10.1111/jcal.12325>
- MacWhinnie, S. G., & Mitchell, C. (2017). English classroom reforms in Japan: A study of Japanese university EFL student anxiety and motivation. *Asian-Pacific Journal of Second and Foreign Language Education*, 2(7), 1–13. <https://doi.org/10.1186/s40862-017-0030-2>
- *Mazzoni, E., & Benvenuti, M. (2015). A robot-partner for preschool children learning English using socio-cognitive conflict. *Educational Technology & Society*, 18(4), 474–485.
- Moussalli, S., & Cardoso, W. (2017). Can you understand me? Speaking robots and accented speech. In K. Borthwick, L. Bradley, & S. Thouësny (Eds.), *CALL in a climate of change: adapting to turbulent global conditions – short papers from EUROCALL 2017* (pp. 217–221). <https://doi.org/10.14705/rpnet.2017.eurocall2017.716>
- Moussalli, S., & Cardoso, W. (2020). Intelligent personal assistants: Can they understand and be understood by accented L2 learners? *Computer Assisted Language Learning*, 33(8), 865–890. <https://doi.org/10.1080/09588221.2019.1595664>

- Nation, I. S. P. (2006). How large a vocabulary is needed for reading and listening? *Canadian Modern Language Review*, 63, 59–82. <https://doi.org/10.3138/cmlr.63.1.59>
- Nor, N. M., & Ab Rashid, R. (2018). A review of theoretical perspectives on language learning and acquisition. *Kasetsart Journal of Social Sciences*, 39(1), 161–167. <https://doi.org/10.1016/j.kjss.2017.12.012>
- Orwin, R. G. (1983). A Fail-SafeN for effect size in meta-analysis. *Journal of Educational Statistics*, 8(2), 157–159. <https://doi.org/10.3102/10769986008002157>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Aki, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., . . . Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *PLOS Medicine*, 18(3), Article e1003583. <https://doi.org/10.1371/journal.pmed.1003583>
- *Park, S. J., Han, J. H., Kang, B. H., & Shin, K. C. (2011). Teaching assistant robot, ROBOSEM, in English class and practical issues for its diffusion. In *Advanced robotics and its social impacts* (pp. 8–11). <https://dx.doi.org/10.1109/ARSO.2011.6301971>
- Pekrun, R., Goetz, T., Titz, W., & Perry, R. P. (2002). Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational Psychologist*, 37, 91–105.
- Pigott, T. D., & Polain, J. R. (2020). Methodological guidance papers: High-quality meta-analysis in a systematic review. *Review of Educational Research*, 90(1), 24–46. <https://doi.org/10.3102%2F0034654319877153>
- Plonsky, L., & Oswald, F. L. (2014). How big is “big”? Interpreting effect sizes in 12 research. *Language Learning*, 64(4), 878–912. <https://doi.org/10.1111/lang.12079>
- Randall, N. (2019). A survey of robot-assisted language learning (RALL). *ACM Transactions on Human-Robot Interaction (THRI)*, 9(1), 1–36. <https://doi.org/10.1145/3345506>
- *Rintjema, E., van den Berghe, R., Kessels, A., de Wit, J., & Vogt, P. (2018). A robot teaching young children a second language: The effect of multiple interactions on engagement and performance. In *HRI'18 Companion: Conference on ACM/IEEE International Conference on Human-Robot Interaction, Chicago, USA*. <https://doi.org/10.1145/3173386.3177059>
- *Saerbeck, M., Schut, T., Bartneck, C., & Janse, M. D. (2010). Expressive robots in education – varying the degree of social supportive behavior of a robotic tutor. In *Proceedings of the 28th ACM Conference on Human Factors in Computing Systems (CHI2010), Atlanta* (pp. 1613–1622). <https://doi.org/10.1145/1753326.1753567>
- *Schodde, T., Hoffmann, L., Stange, S., & Kopp, S. (2019). Adapt, explain, engage—A study on how social robots can scaffold second-language learning of children. *ACM Transactions on Human-Robot Interaction*, 9(1), 1–27. <https://doi.org/10.1145/3366422>
- Sheen, Y. (2008). Recasts, language anxiety, modified output, and L2 learning. *Language Learning*, 58(4), 835–874.
- *Shin, J., & Shin, D. (2015). *Robot as a facilitator in language conversation class*. Presented at Tenth Annual ACM/IEEE International Conference on Human-Robot. <https://doi.org/10.1145/2701973.2702062>

- Singleton, D., & Leśniewska, J. (2021). The critical period hypothesis for L2 acquisition: An unfalsifiable embarrassment?. *Languages*, 6(3), 149–164. <https://doi.org/10.3390/languages6030149>.
- Tai, T.-Y., Chen, H. H.-J., & Todd, G. (2020). The impact of a virtual reality app on adolescent EFL learners' vocabulary learning. *Computer Assisted Language Learning*. Advance online publication. <https://doi.org/10.1080/09588221.2020.1752735>
- Teng, M. F. (2021). Interactive-whiteboard-technology-supported collaborative writing: Writing achievement, metacognitive activities, and co-regulation patterns. *System*. Advance online publication. <https://doi.org/10.1016/j.system.2020.102426>
- Teo, T., Khazaei, S., & Derakhshan, A. (2022). Exploring teacher immediacy-(non) dependency in the tutored augmented reality game-assisted flipped classrooms of English for medical purposes comprehension among the Asian students. *Computers and Education*, 179. <https://doi.org/10.1016/j.compedu.2021.104406>
- Torgerson, C. J. (2006). Publication bias: The Achilles' heel of systematic reviews? *British Journal of Educational Studies*, 54, 89–102. <https://doi.org/10.1111/j.1467-8527.2006.00332.x>
- *van den Berghe, R., de Haas, M., Oudgenoeg-Paz, O., Krahmer, E., Verhagen, J., Vogt, P., Willemsen, B., de Wit, J., & Leseman, P. (2021). A toy or a friend? Children's anthropomorphic beliefs about robots and how these relate to second-language word learning. *Journal of Computer Assisted Language Learning*, 37(2), 396–410. <https://doi.org/10.1111/jcal.12497>
- van den Berghe, R., Verhagen, J., Oudgenoeg-Paz, O., van der Ven, S., & Leseman, P. (2019). Social robots for language learning: A review. *Review of Educational Research*, 89(2), 259–295. <https://doi.org/10.3102%2F0034654318821286>
- *Van Minkelen, P., Gruson, C., van Hees, P., Willems, M., de Wit, J., Aarts, R., Denissen, J., & Vogt, P. (2020). Using self-determination theory in social robots to increase motivation in L2 word learning. In *Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI '20)* (pp. 1–9). <https://doi.org/10.1145/3319502.3374828>
- Viechtbauer, W., & Cheung, M. W.-L. (2010). Outlier and influence diagnostics for meta-analysis. *Research Synthesis Methods*, 1(2), 112–125. <https://doi.org/https://doi.org/10.1002/jrsm.11>
- *Vogt, P., van den Berghe, R., De Haas, M., Hoffman, L., Kanero, J., Mamus, E., Montanier, J.-M., Oranc, C., Oudgenoeg-Paz, O., Garcia, D. H., Papadopoulos, F., Schodde, T., Verhagen, J., Wallbridge, C. D., Willemsen, B., de Wit, J., Belpaeme, T., Göksun, T., Kopp, S., . . . Pandey, A. K. (2019, March). Second language tutoring using social robots: A large-scale study. In *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 497–505). IEEE. <https://doi.org/10.1109/HRI.2019.8673077>
- *Wedenborn, A., Wik, P., Engwall, O., & Beskow, J. (2019). The effect of a physical robot on vocabulary learning. *International Workshop on Spoken Dialogue Systems*. <https://doi.org/10.48550/arXiv.1901.10461>
- *Westlund, J. K., Dickens, L., Jeong, S., Harris, P., DeSteno, D., & Breazeal, C. (2015). A Comparison of children learning new words from robots, tablets, & people. *Proceedings of New Friends: The 1st International Conference on Social Robots in Therapy and Education, The Netherlands*. Retrieved January 25, 2021, from <https://www.media.mit.edu/publications/a-comparison-of-children-learning-from-robots-tablets-and-people/>.

- Wu, M.-H. (2021). The applications and effects of learning English through augmented reality: A case study of Pokémon Go. *Computer Assisted Language Learning*, 34(5–6), 778–812. <https://doi.org/10.1080/09588221.2019.1642211>
- *Wu, W.-C. V., Wang, R.-J., & Chen, N.-S. (2015). Instructional design using an in-house built teaching assistant robot to enhance elementary school English-as-a-foreign-language learning. *Interactive Learning Environments*, 23(6), 696–714. <https://doi.org/10.1080/10494820.2013.792844>
- Xie, Y., Chen, Y., & Ryder, L. H. (2021). Effects of using mobile-based virtual reality on Chinese L2 students' oral proficiency. *Computer Assisted Language Learning*, 34(3), 225–245. <https://doi.org/10.1080/09588221.2019.1604551>
- Yin, Q., & Satar, M. (2020). English as a foreign language learner interaction with chatbots: Negotiation for meaning. *International Online Journal of Education and Teaching (IOJET)*, 7(2), 390–410. <https://iojet.org/index.php/IOJET/article/view/707>
- Zeng, C., Li, S., Li, Q., Hu, J., & Hu, J. (2020). A survey on machine reading comprehension: Tasks, evaluation metrics and benchmark datasets. *Applied Sciences*, 10(21), 7640. <https://doi.org/10.3390/app10217640>
- Zhang, S., & Zhang, X. (2022). The relationship between vocabulary knowledge and L2 reading/listening comprehension: A meta-analysis. *Language Teaching Research*, 26(4), 696–725. <https://doi.org/10.1177%2F1362168820913998>

Authors

ALI DERAKHSHAN is a professor of applied linguistics at the English Language and Literature Department, Faculty of Humanities and Social Sciences, Shahid Beheshti Campus, Golestan University, Shahid Beheshti St, P. O. Box: 155, Gorgan, Golestan 49138 15759, Iran; e-mail: a.derakhshan@gu.ac.ir. His research interests are positive psychology, teacher education, learner-individual differences, technology and language education, cross-cultural interpersonal factors in educational psychology, and intercultural communication.

TIMOTHY TEO is a professor of education at The Chinese University of Hong Kong; e-mail: timothy/teo@cuhk.edu.hk. His research interests are multidisciplinary and include both substantive and methodological areas. These are educational psychology (self-efficacy-teachers and students; beliefs about teaching and learning; meta-cognition), ICT in education (technology acceptance and adoption; e-learning), music education (psychological processes of music teaching and learning), and quantitative methods (psychometrics; instrument development and validation; cross-cultural measurement; issues in survey development and administration; structural equation modeling; multilevel modeling; latent growth modeling).

ESMAEEL SAEEDY ROBAT is an assistant professor of literacy and language education at the Department of Education, Islamic Azad University, Taybad Branch, Iran; e-mail: es.saeedy@iau.ac.ir. His research interests include educational neuroscience, technology-aided education, and meta-analytic research.

MOSTAFA JANEHI ENAYAT is an assistant professor of applied linguistics at the English Department, University of Maragheh, Maragheh, Iran; e-mail: m.enayat@maragheh.ac.ir. His areas of interest include language testing, vocabulary instruction and assessment, educational technology, and critical discourse analysis.

AKBAR A. JAHANABKSH holds a PhD in applied linguistics (TEFL) and has been working as an EFL instructor, freelance researcher, and statistical analyst at the Bureau of Education at Tehran's 19th District, Ministry of Education, Iran; e-mail: *Jahaanbakhsh2@gmail.com*. His main areas of interest are research methodology, methodological rigor and study quality, advanced quantitative analyses, and meta-analytic research.